Automating QoS–QoE Translation with Agentic Systems

Yingjie Yu, Mingyuan Wu, Ahmadreza Eslaminia, Lingzhi Zhao, Kaizhuo Yan, and Klara Nahrstedt
University of Illinois Urbana-Champaign, Urbana, Illinois, USA
{yyu69,mw34,ae15,lz26,kaizhuo2,klara}@illinois.edu

***Abstract*—Video streaming involves two dimensions of quality: measurable quality-of-service (QoS) parameters, such as bandwidth and latency, and user-perceived quality-of-experience (QoE), commonly assessed using mean opinion score (MOS). Automating bidirectional translation between these dimensions is important for predicting user experience from system conditions and identifying system requirements for a desired experience. However, published QoS–QoE relationships vary in parameter definitions, contextual dependencies, and reporting formats, hindering automated reuse. To address these challenges, we make three contributions. First, we develop a unified, source-grounded QoS–QoE data schema that represents heterogeneous relationships, parameter definitions, supporting evidence, and contextual factors. Second, we construct a literature-derived dataset through an agentic, human-guided workflow integrating paper curation, relationship extraction, metadata enrichment, and feedback-driven multi-reviewer quality evaluation, yielding 1,026 relationship records from 505 video-streaming papers. Third, we evaluate bidirectional QoS–QoE translation by fine-tuning five open-source large language models (LLMs) on the constructed dataset and assessing their performance on QoS → QoE and QoE → QoS structured prediction tasks. The best overall post-fine-tuning results achieve 8.49% mean absolute percentage error on continuous targets and 90.24% accuracy on discrete labels. While our evaluation focuses on video streaming, the proposed framework provides a foundation for extending automated QoS–QoE translation to other multimedia services. The complete dataset and code are publicly available at https://yyu6969.github.io/qos-qoe-translation-page/.**

***Index Terms*—Quality of Service, Quality of Experience, Large Language Model, Video Streaming, Multimedia Systems**

I. Introduction

Video-streaming quality has two closely related dimensions: Quality of Service (QoS) and Quality of Experience (QoE). QoS describes measurable system and network conditions, including bandwidth, latency, bitrate, and packet loss, while QoE reflects users' perceived quality, often assessed using metrics such as mean opinion score (MOS) [1], [2]. Developers and service providers manage QoS parameters to deliver reliable streaming, whereas users prioritize viewing experience. Understanding the relationship between system conditions and perceived quality is essential for streaming design and management [3], [4].

This relationship motivates automated bidirectional QoS-QoE translation. For QoS→QoE, service providers and system developers may predict user-perceived quality from measurable conditions, including available bandwidth, latency, and rebuffering behavior, to support quality monitoring, system configuration, and adaptive video streaming without new subjective user studies for every operating condition. For QoE→QoS, users or system designers may specify a desired viewing experience and identify suitable network and system conditions, including bandwidth requirements and streaming configurations, to facilitate resource planning and QoE-aware system design without manually deriving complex QoS-QoE mappings. Beyond prediction, a structured, source-grounded representation could support evidence-based querying and cross-study comparison. However, QoS-QoE relationships are context-dependent and not necessarily one-to-one: multiple QoS parameters can jointly affect QoE, while similar QoE outcomes may arise from different system configurations, streaming protocols, devices, and experimental settings. Bidirectional translation therefore requires jointly representing system conditions, user-perceived quality, and contextual dependencies.

Prior work has investigated QoS-QoE relationships and quality-aware video streaming through subjective experiments, analytical models, and learning-based approaches, examining how bitrate adaptation, stalling, startup delay, and network impairments affect perceived quality [2], [4]–[6]. Existing QoE datasets provide resources for studying specific experimental environments and measurement settings [7], [8]. However, published QoS-QoE findings remain scattered across scientific papers with heterogeneous parameter definitions, contextual conditions, and reporting formats, including text, tables, figures, and equations, complicating systematic extraction, comparison, and reuse. Despite existing models and empirical findings, these relationships are not readily available in a unified, source-grounded form supporting automated bidirectional translation across heterogeneous video-streaming scenarios.

Addressing these limitations presents three key challenges. First, a unified data schema must represent heterogeneous QoS-QoE relationships involving multiple QoS and QoE parameters, capturing parameter definitions, relationship expressions, supporting evidence, contextual dependencies, and the experimental conditions under which they were established. Second, constructing a source-grounded dataset requires identifying relevant scientific papers, extracting relationships from diverse document structures, enriching contextual metadata, and evaluating the reliability of extracted records. Third, validating the dataset's utility for automated bidirectional translation requires determining whether large language models (LLMs) trained on these structured relationships can predict QoE from QoS conditions and infer suitable QoS conditions from desired QoE outcomes.

To address these challenges, we develop *QoS-QoE Translation* with three main contributions: **(1) Unified QoS-QoE Data Schema:** a source-grounded representation of heterogeneous QoS-QoE relationships, parameter definitions, relationship expressions, supporting evidence, and contextual metadata covering streaming protocols, network conditions, devices, user preferences, and experimental scenarios; **(2) Agentic Dataset Construction Framework:** an agentic framework integrating AI-assisted paper curation, source-grounded LLM extraction, metadata enrichment, and feedback-driven multi-reviewer quality evaluation, producing 1,026 relationship records from 505 video-streaming papers; and **(3) Bidirectional QoS-QoE Translation and Evaluation:** fine-tuning and evaluating five open-source LLMs for structured QoS-to-QoE and QoE-to-QoS prediction, achieving best post-fine-tuning results of 8.49% mean absolute percentage error on continuous targets and 90.24% accuracy on discrete labels. While evaluated on video streaming, the proposed schema and framework provide a foundation for other multimedia services.

II. Related Work

QoS-QoE Modeling and System Adaptation. Multimedia research has long examined the relationship between application-level quality requirements and system-level resources. Early work, such as the QoS Broker, studied QoS translation and resource management for distributed multimedia applications [9]. Subsequent studies examined how network impairments, bitrate adaptation, startup delay, and rebuffering affect perceived quality in video streaming, relating measurable QoS parameters to user-perceived QoE through subjective experiments, analytical models, and learning-based methods [2]–[4]. Adaptive streaming approaches explored how system conditions and quality objectives guide streaming decisions [5], [6]. In contrast, our work systematically represents and reuses QoS-QoE relationships across heterogeneous studies to support automated bidirectional translation.

QoE Datasets and Evaluation Resources. Existing QoE datasets support understanding and predicting user-perceived quality and evaluating QoE models under specific measurement settings. The Waterloo Streaming QoE Database provides subjective assessments of adaptive video streams across varying content, adaptation strategies, and network conditions [7], while SNESet provides QoS and QoE measurements in edge-hosted video streaming applications [8]. Rather than collecting new subjective observations, our work consolidates published QoS-QoE relationships from scientific papers into a unified, source-grounded representation, preserving parameter definitions, supporting evidence, and contextual metadata for cross-study comparison and reuse.

LLM-Assisted Scientific Document Extraction. Recent research has investigated the use of LLMs for structured information extraction from scientific literature [10], [11]. Layout-aware models support processing complex document structures, including tables and other structured content [12]. Iterative refinement and LLM-based evaluation have been explored to improve output quality through feedback and review [13], [14]. Building on these capabilities, our framework constructs a literature-derived QoS-QoE dataset through source-grounded extraction, contextual metadata enrichment, and iterative multi-reviewer evaluation, populating a unified QoS-QoE relationship schema for downstream bidirectional translation rather than general-purpose document extraction.

III. QoS-QoE Relationships and Data Representation

A. QoS-QoE Relationships in Video Streaming

In video streaming, QoS characterizes measurable network and application conditions, including bandwidth, bitrate, latency, packet loss, startup delay, and rebuffering behavior. QoE describes user-perceived service quality, commonly measured through subjective assessments such as MOS [1], [2]. MOS aggregates subjective ratings, although its numerical scale and interpretation may differ across studies. Other studies report objective quality metrics, such as VMAF, PSNR, and SSIM, as quality-related outcomes; these metrics are not necessarily interchangeable with subjective QoE measurements, making their original definitions important for interpreting reported QoS-QoE relationships.

QoS-QoE relationships are often complex and not necessarily one-to-one: multiple QoS parameters may jointly influence a single QoE outcome, while similar QoE values may arise from different combinations of system conditions. Figure 1 presents an example of a source-reported QoS-QoE relationship, $\mathrm{MOS} = 4.23 - 0.0672L_{ti} - 0.742L_{fr} - 0.106L_{tr}$, where L_{ti}, L_{fr}, and L_{tr} represent impairment levels of initial loading delay, rebuffering frequency, and average rebuffering duration, respectively. Each variable uses an ordinal scale from 1 (low) to 3 (high), whereas MOS commonly uses a five-point subjective rating scale from 1 (bad) to 5 (excellent), although scales may vary across studies. Holding other variables constant, the negative coefficients indicate that higher impairment levels are associated with lower predicted MOS. This example highlights why preserving parameter definitions and measurement scales is essential for interpreting multi-parameter QoS-QoE relationships.

Furthermore, QoS-QoE relationships depend on the conditions under which they are observed. Streaming protocols, network types, client devices, video categories, and experimental scenarios may influence how system conditions relate to perceived quality. For example, DASH-based adaptive video-on-demand and WebRTC-based interactive streaming may involve different latency, rebuffering, and quality requirements. User preferences and historical observations can provide additional context when reported. Therefore, a unified representation must capture not only QoS and QoE parameters and their relationships, but also the contextual dependencies and source evidence needed for consistent interpretation across studies.

B. Unified QoS-QoE Data Schema

To represent heterogeneous QoS-QoE relationships, we develop a unified, source-grounded JSON data schema. Each



Fig. 1: An example source-grounded QoS-QoE relationship record with contextual metadata and extracted relationship.

TABLE I: Field definitions of QoS-QoE Translation.

Field	Definition
id	Unique record identifier.
year	Source publication year.
venue	Source publication venue.
domain	Application domain.
protocol	Streaming or transport protocol.
network_type	Access network type.
device_type	Client device type.
video_type	Video category or format.
user_preference	User preference in the study.
scenario	Experimental or evaluation setting.
history_log	Time-ordered QoS-QoE observations.
data_type	Evidence type (equation, table, figure).
qos_parameter	QoS variables in the relationship.
qos_parameter_definition	Definitions of QoS variables.
qoe_parameter	QoE variables in the relationship.
qoe_parameter_definition	Definitions of QoE variables.
relationship	Extracted QoS-QoE dependency.
description	Natural-language explanation.
source	Traceable source evidence.

record conceptually integrates two main components: **relationship**, which describes the reported QoS-QoE dependency, and **metadata**, which preserves source information and contextual conditions. This design accommodates relationships involving multiple QoS and QoE parameters while retaining their original definitions and supporting evidence. Figure 1 illustrates an example record, and Table I summarizes the schema fields.

The relationship representation includes `qos_parameter` and `qoe_parameter`, which identify the variables involved in a reported dependency, together with `qos_parameter_definition` and `qoe_parameter_definition`, which preserve their original meanings. The `relationship` field stores the extracted dependency, while `description` provides a natural-language explanation. The `data_type` and `source` fields identify the evidence format and supporting source, enabling the relationship to be traced back to the original literature.

To capture contextual dependencies, the schema includes `protocol`, `network_type`, `device_type`, `video_type`, `user_preference`, and `scenario`. These fields describe the streaming environment and experimental conditions associated with a relationship. The `history_log` field preserves time-ordered QoS-QoE observations when available, while publication metadata, including `year` and `venue`, identifies the originating study.

Together, the relationship and metadata fields link multi-parameter QoS-QoE dependencies to their experimental conditions, observation history, and source evidence, enabling traceability and cross-study comparison. The following section describes how the agentic dataset construction framework populates this schema from scientific literature.

IV. Agentic Dataset Construction Framework

Building on the unified QoS-QoE data schema introduced in Section III, we develop an agentic framework for constructing a source-grounded dataset from scientific literature. The framework organizes task-specific LLM components to identify QoS-QoE relationships across heterogeneous research papers and transform the reported parameters, relationships, and contextual information into structured records conforming to the proposed schema. Figure 2 illustrates the construction workflow, which consists of three stages: AI-assisted paper curation with human screening, QoS-QoE relationship extraction with metadata enrichment, and iterative multi-reviewer data evaluation. These stages progressively populate the proposed schema, while reviewer feedback and human-guided re-evaluation enable refinement of extracted records. Although the current dataset focuses on video streaming, the framework can be adapted to other domains requiring source-grounded relationship extraction from scientific literature.

A. Paper Curation

We begin by constructing a curated corpus of research papers reporting QoS-QoE relationships in video-streaming. To identify relevant studies, we combine human screening with OpenAI Deep Research-assisted literature discovery [15], selecting papers containing extractable QoS-QoE relationships, supporting evidence, and sufficient contextual information. Duplicate papers are removed through manual screening.

We focus on papers published between 2017 and 2025, as advances in streaming systems, codecs, devices, and network configurations can affect the applicability of reported QoS-QoE relationships. The collected literature spans different video-streaming environments, including DASH, HTTP-based streaming, WebRTC, and other streaming or transport protocols. These studies provide relationships observed under diverse network, device, and application conditions, which are important for populating the contextual fields of the schema.

Through this human-AI curation process, we collect 505 video-streaming research papers. The curated corpus serves as the source material for subsequent relationship extraction and

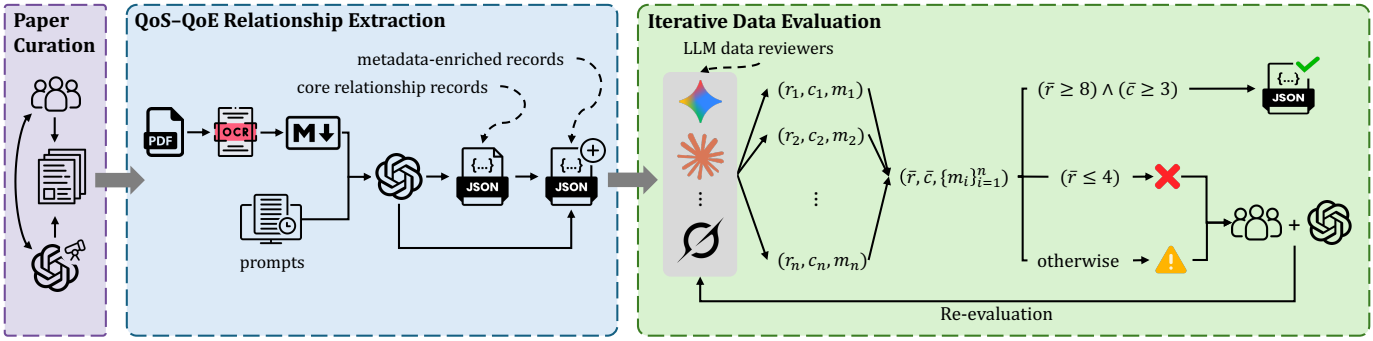


Fig. 2: Overview of the QoS-QoE Translation dataset construction pipeline. The pipeline begins with paper curation, followed by QoS-QoE relationship extraction and iterative data evaluation.

metadata enrichment. The distribution of venues and streaming environments is examined in the evaluation section.

B. QoS-QoE Relationship Extraction

We extract QoS-QoE relationships and supporting information from the curated PDF papers. MinerU [16], a document parser with OCR capabilities, converts the PDFs into machine-readable Markdown while preserving content and document structure. The converted documents and task-specific prompts are provided to an LLM for structured extraction.

We use GPT-5.2 Thinking [17] as the core extraction model, leveraging its reasoning and long-context capabilities to process complex academic papers. Guided by the unified data schema, it identifies QoS and QoE parameters, their definitions, reported relationships, and supporting evidence from equations, tables, and figures.

Extraction produces two levels of records. First, core relationship records capture source-grounded QoS-QoE dependencies and associated parameter definitions. Second, metadata enrichment adds contextual attributes, including streaming protocol, network type, device type, video type, user preferences, and experimental scenario. Missing contextual fields are inferred from surrounding document content only when supported by source evidence.

Separating relationship extraction from metadata enrichment distinguishes reported dependencies from contextual information. Together, these steps populate the relationship and metadata fields defined in Section III, producing structured, source-grounded records for downstream evaluation and reuse.

C. Iterative Data Evaluation

To improve the quality of the constructed dataset, we introduce an iterative data evaluation stage. As shown in Figure 2, each metadata-enriched record is reviewed by multiple LLM-based data evaluators. In our dataset construction setup, we use three reviewers: Gemini-2.5-flash-lite [18], Claude-haiku-4-5-20251001 [19], and Grok-4.20-0309-reasoning [20].

Each reviewer provides a rating, a confidence score, and written feedback, represented as (r_i, c_i, m_i). The rating r_i uses a 0–10 scale, where 0, 2, 4, 6, 8, and 10 correspond to strong reject, reject, weak reject, weak accept, accept, and strong accept, respectively. The confidence score c_i ranges from 1 to 5, indicating the reviewer's confidence in its assessment of the extracted record. The feedback m_i describes identified issues and suggests possible revisions.

The reviewer outputs are aggregated using the average rating $\bar{r}$, average confidence $\bar{c}$, and collected feedback. Based on empirical inspection of reviewer outputs during dataset construction, we use the following decision thresholds to balance the retention of high-quality records against the filtering of unreliable extractions:

$$\text{Decision} = \begin{cases} \textbf{Accept}, & \text{if } \bar{r} \geq 8 \text{ and } \bar{c} \geq 3, \\ \textbf{Reject}, & \text{if } \bar{r} \leq 4, \\ \textbf{Conditional Accept}, & \text{otherwise.} \end{cases} \tag{1}$$

Records satisfying the accept condition are retained as JSON entries, while rejected and conditionally accepted records are forwarded to a re-evaluation stage. During re-evaluation, human guidance and an LLM are jointly used to revise the records based on reviewer feedback before returning them to the evaluation loop. This iterative mechanism aims to identify and address unsupported relationships, ambiguous parameter definitions, and inconsistent contextual information.

The reviewer ratings and confidence scores are used to assess the quality of extracted records rather than the predictive performance of downstream QoS-QoE translation models. We examine the reliability of this quality-control process through independent human audits and reviewer agreement analysis.

D. Human Audit and Reviewer Agreement

To independently assess the quality of the LLM-assisted construction process, two annotators audit 50 uniformly sampled items from each of three stages: candidate papers, re-evaluated records, and final dataset records. Table II reports three complementary metrics: the unanimous positive rate, which requires both annotators to judge an item valid; the criterion-level pass rate, which aggregates all applicable criterion judgments; and the agreement rate, which measures identical judgments between annotators.

Across the three audit targets, criterion-level pass rates range from 91.6% to 98.0%, and unanimous positive rates from

TABLE II: Two-annotator human audit results by stage (%).

Audit target	Unanimous positive rate	Criterion-level pass rate	Agreement rate
Paper curation	94.0	98.0	94.0
Re-evaluated records	72.0	96.4	78.0
Final dataset records	74.0	91.6	78.0

72.0% to 94.0%. The three LLM reviewers achieve 73.49% unanimous agreement on binary reject versus non-reject decisions across all 1,026 records, with pairwise agreement of 78.95%–86.75%. These results characterize the reliability and limitations of the construction quality-control process, but reviewer agreement alone does not establish the correctness of every extracted record. Appendix F provides detailed sampling protocols, stage-specific criteria, reviewer-comment analysis, audit results, and error analysis.

Through paper curation, relationship extraction, metadata enrichment, and iterative evaluation, the framework constructs 1,026 structured QoS-QoE relationship records from 505 curated papers. The resulting dataset is then assessed for its coverage and utility in bidirectional QoS-QoE translation.

V. Evaluation

We evaluate *QoS-QoE Translation* from two perspectives: the coverage of its source-grounded relationships across video-streaming studies, and its utility for automated bidirectional QoS-QoE translation. For the latter, we fine-tune independent prediction LLMs on the constructed dataset, rather than reusing the LLMs employed for extraction and record evaluation. We report MAPE and Accuracy@δ for continuous values and Accuracy and Macro-F1 for discrete labels, comparing performance before and after supervised fine-tuning (SFT) and across both translation directions.

A. Dataset Sources and Characteristics

The constructed dataset contains 1,026 QoS-QoE relationship records from 505 curated video-streaming papers. Figure 3 characterizes the represented streaming environments, publication years and venues, and QoS/QoE parameters. These distributions describe the dataset's scope and do not by themselves establish equal coverage of all operating conditions.

Streaming environments. Figures 3a–3d summarize the represented streaming contexts. DASH accounts for 59.7% of protocol entries and HTTP for 20.2%, while WebRTC, QUIC, RTP, and HLS are also represented. The largest network categories are cellular 4G (33.2%), wired (23.0%), and Internet-based environments (19.3%). Desktop (45.2%) and mobile (41.0%) are the most common device categories; 2D video-on-demand (61.3%) and 2D live streaming (27.8%) dominate video type. The data thus covers multiple video-streaming environments but is concentrated in common settings.

Publication sources. Figure 3e shows coverage from 2017 to 2025, with 2024 accounting for the largest share (18.2%). Figure 3f reports sources from multimedia and networking venues, led by ACM MM (26.0%), arXiv (16.7%), IEEE Access (15.6%), IEEE INFOCOM (9.5%), ACM TOMM (8.2%), and NSDI (7.6%). These distributions document the provenance and temporal scope of the collected literature.

QoS and QoE parameters. Figures 3g and 3h show the represented parameters. Bitrate is the most frequent QoS parameter (44.1%), followed by rebuffering time (16.2%), with rebuffering duration, jitter, buffer occupancy, bandwidth, CPU usage, packet loss rate, and startup delay also represented. On the QoE side, QoE score (33.5%), MOS (25.4%), and VMAF (17.2%) dominate; PSNR, VISQOL, SSIM, and other quality-related measures also appear. Some variables, including rebuffering time, occur on both sides according to their definitions in the source studies. This heterogeneity motivates preserving study-specific parameter definitions in the schema.

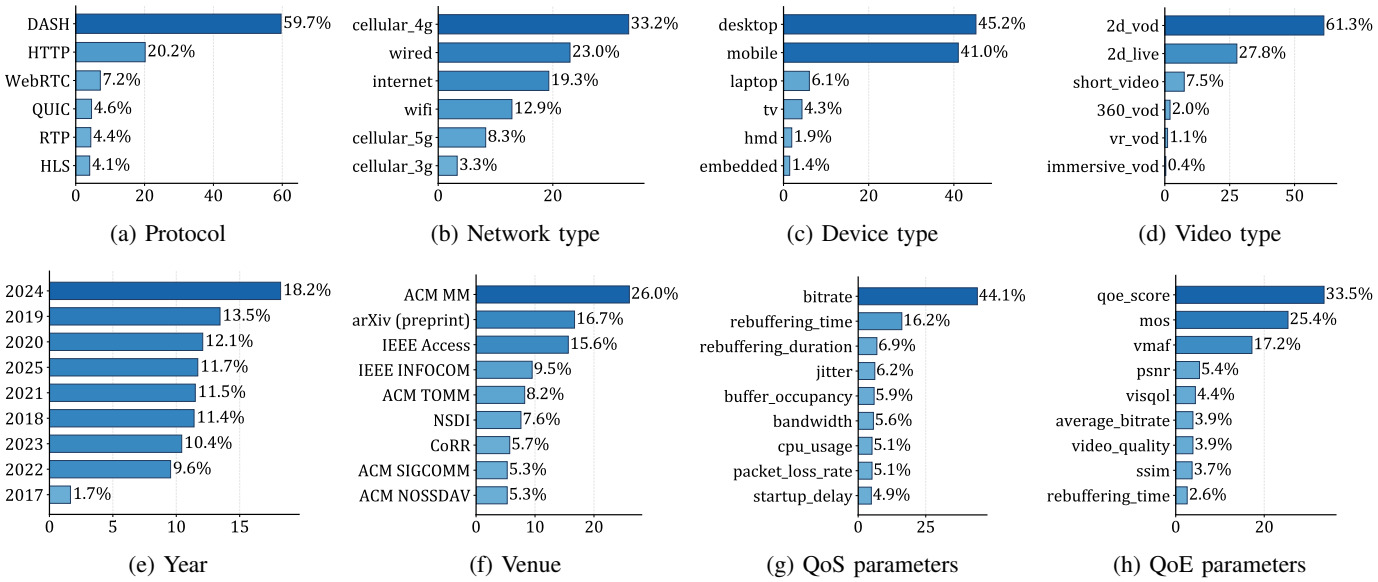
B. Translation Evaluation Protocol and Metrics

Task formulation. To validate the dataset's utility, we formulate QoS→QoE and QoE→QoS as structured prediction tasks derived from source-grounded relationship records. The former predicts user-experience parameters from system-level conditions, while the latter predicts system-level conditions from user-experience observations. Both use structured input instances with contextual information and output the queried target field in JSON. For example, using the record in Figure 1, QoS-to-QoE translation predicts MOS from the specified initial loading delay and rebuffering impairment levels, while the reverse task predicts a queried QoS impairment level from MOS and the available context. These prediction models are distinct from the LLMs used to construct and review the records in Section IV.

Supervised fine-tuning. We evaluate Qwen3-8B, Qwen3-32B, Qwen3.5-35B-A3B, Llama-3.1-8B-Instruct, and Llama-3.3-70B-Instruct [21]–[23]. Using Tinker [24], we fine-tune these models on chat-formatted JSON instances with an instruction, task identifier, contextual metadata, scenario description, parameter mapping, source-grounded evidence, history log, and query; each target contains only the predicted JSON field.

Before instance construction, we split 1,026 source-grounded relationship records at the source-record level into 913 training and 113 testing records, preventing instances from the same record from crossing splits. For each instance, holdout-based history reconstruction removes one time point from the history log, using its observed side as the query and the corresponding target side as the output. This yields 8,107 instances: 7,205 for training and 902 for testing. A leakage audit finds zero overlap between splits at the source-record, record-task-query, and normalized exact task-query-target levels (Appendix C). These checks do not establish a paper-level holdout. All models use a maximum sequence length of 32,768 tokens, batch size 128, learning rate 2×10^{-4}, a linear learning-rate schedule, and one training epoch.

Metrics. For continuous-value prediction, we report mean absolute percentage error (MAPE; lower is better) and Accuracy@δ (higher is better), the fraction of predictions within a parameter-specific absolute or relative tolerance. The tolerance configuration is provided in Appendix D. For



(a) Protocol (b) Network type (c) Device type (d) Video type

(e) Year (f) Venue (g) QoS parameters (h) QoE parameters

Fig. 3: Dataset analysis of QoS-QoE Translation. The first row shows distributions over protocol, network type, device type, and video type. The second row shows distributions over publication year, venue, and QoS/QoE parameters.

discrete-label prediction, we report exact-match Accuracy and Macro-F1, which weights classes equally.

C. Overall Translation Performance

Table III compares the five models before and after SFT. All models improve on the reported continuous-value and discrete-label metrics, indicating that the dataset provides a useful supervision signal for the structured translation tasks. Qwen3.5-35B-A3B achieves the strongest overall post-SFT results: 8.49% MAPE, 83.41% Accuracy@δ, 90.24% discrete-label Accuracy (tied with Llama-3.3-70B-Instruct), and 84.63% Macro-F1. For example, its MAPE decreases from 14.43% to 8.49%, while discrete-label Accuracy increases from 69.92% to 90.24%. The results support the dataset's utility for fine-tuning prediction models, rather than establishing performance in untested deployment environments.

D. Bidirectional Translation Analysis

Table IV reports post-SFT performance by direction. For QoS→QoE, Llama-3.3-70B-Instruct achieves the lowest MAPE (6.88%), ties for highest Accuracy@δ (83.77%), and leads in discrete-label Accuracy (85.00%) and Macro-F1 (77.78%). For QoE→QoS, Qwen3.5-35B-A3B leads across all metrics: MAPE (9.40%), Accuracy@δ (83.02%), discrete-label Accuracy (96.39%), and Macro-F1 (91.81%).

Across the evaluated models, reverse QoE→QoS translation has higher continuous-value errors than forward translation, while its discrete-label performance is often stronger. A possible explanation is that similar QoE values may arise from multiple QoS configurations, making exact reverse numeric prediction less constrained. These results demonstrate predictive utility in both directions on the held-out instances, but do not establish a unique or operationally optimal QoS configuration for a target QoE.

VI. Discussion and Future Work

Our results demonstrate the utility of literature derived QoS-QoE relationships for training bidirectional translation models, as shown in Tables III and IV. However, performance on held-out prediction instances does not establish the operational validity of predicted system configurations, as multiple QoS conditions may correspond to similar QoE outcomes.

Several limitations remain. As shown in Figure 3, the dataset is concentrated in commonly studied streaming environments and parameters, potentially limiting its coverage of less-represented scenarios. Although iterative multi-reviewer evaluation and human audits improve quality control, extraction correctness is not guaranteed for every record. Furthermore, our source-record-level split does not establish generalization to entirely unseen papers, and the current experiments do not evaluate performance in real-world streaming deployments.

Future work will focus on expanding dataset coverage, strengthening human validation, and evaluating cross-paper generalization and practical deployment. Beyond structured prediction, the source-grounded schema could support evidence-based retrieval, cross-study comparison, and LLM-based assistants for interpreting QoS-QoE relationships. Extending the framework to other multimedia services, such as teleconferencing, is another promising direction.

VII. Conclusion

We present *QoS-QoE Translation*, integrating a unified, source-grounded data schema, an agentic dataset construction framework, and bidirectional QoS-QoE translation evaluation. The schema captures heterogeneous QoS-QoE relationships, parameter definitions, contextual metadata, and supporting evidence. Through paper curation, relationship extraction, metadata enrichment, and iterative quality evaluation, the frame-

TABLE III: Overall model evaluation results before and after SFT. All values are percentages.

Model	Before SFT				After SFT			
	MAPE↓	Accuracy@δ↑	Accuracy↑	Macro-F1↑	MAPE↓	Accuracy@δ↑	Accuracy↑	Macro-F1↑
Qwen3-8B	20.23	64.20	67.48	55.63	11.79	78.13	80.49	72.29
Qwen3-32B	16.46	67.95	68.29	56.69	9.41	80.24	84.55	70.12
Qwen3.5-35B-A3B	14.43	**72.94**	69.92	**61.33**	**8.49**	**83.41**	**90.24**	**84.63**
Llama-3.1-8B-Instruct	26.72	55.90	50.41	35.62	11.34	79.49	83.74	70.79
Llama-3.3-70B-Instruct	**13.76**	72.70	**70.73**	56.29	9.49	81.90	**90.24**	81.93

TABLE IV: Overall model evaluation results after SFT for QoS→QoE and QoE→QoS translation. All values are percentages.

Task	Model	MAPE (%)↓	Accuracy@δ (%)↑	Accuracy (%)↑	Macro-F1 (%)↑
QoS → QoE	Qwen3-8B	9.11	78.55	75.00	73.43
	Qwen3-32B	7.41	82.61	77.50	62.18
	Qwen3.5-35B-A3B	7.05	**83.77**	77.50	66.53
	Llama-3.1-8B-Instruct	8.79	81.74	75.00	61.31
	Llama-3.3-70B-Instruct	**6.88**	**83.77**	**85.00**	**77.78**
QoE → QoS	Qwen3-8B	13.47	77.67	83.13	71.43
	Qwen3-32B	10.66	77.67	87.95	73.45
	Qwen3.5-35B-A3B	**9.40**	**83.02**	**96.39**	**91.81**
	Llama-3.1-8B-Instruct	12.94	77.04	87.95	73.79
	Llama-3.3-70B-Instruct	11.12	79.87	92.77	82.74

work constructs 1,026 relationship records from 505 video-streaming papers. Experiments with five open-source LLMs demonstrate the dataset's utility for structured QoS-to-QoE and QoE-to-QoS prediction, with supervised fine-tuning improving both directions. These results provide a foundation for automated, source-grounded QoS-QoE translation research.

REFERENCES

[1] ITU-T, "Roadmap for qos and qoe in the itu-t study group 12 context," International Telecommunication Union, Tech. Rep. GSTR-RQ, Sep. 2023. [Online]. Available: https://www.itu.int/dms_pub/itu-t/opb/tut/T-TUT-QOS-2023-2-PDF-E.pdf

[2] M. Alreshoodi and J. Woods, "Survey on qoe/qos correlation models for multimedia services," *International Journal of Distributed and Parallel Systems*, vol. 4, no. 3, pp. 53–72, 2013. [Online]. Available: https://doi.org/10.5121/ijdps.2013.4305

[3] A. A. Barakabitze *et al.*, "Qoe management of multimedia streaming services in future networks: A tutorial and survey," *IEEE Communications Surveys & Tutorials*, vol. 22, no. 1, pp. 526–565, 2020.

[4] N. Barman and M. G. Martini, "Qoe modeling for http adaptive video streaming: A survey and open challenges," *IEEE Access*, vol. 7, pp. 30 831–30 859, 2019. [Online]. Available: https://doi.org/10.1109/ACCESS.2019.2901778

[5] H. Mao, R. Netravali, and M. Alizadeh, "Neural adaptive video streaming with pensieve," in *Proceedings of the Conference of the ACM Special Interest Group on Data Communication*, ser. SIGCOMM '17. New York, NY, USA: Association for Computing Machinery, 2017, pp. 197–210.

[6] X. Yin, A. Jindal, V. Sekar, and B. Sinopoli, "A control-theoretic approach for dynamic adaptive video streaming over http," in *Proceedings of the 2015 ACM Conference on Special Interest Group on Data Communication*, ser. SIGCOMM '15. New York, NY, USA: Association for Computing Machinery, 2015, pp. 325–338.

[7] Z. Duanmu, A. Rehman, and Z. Wang, "A quality-of-experience database for adaptive video streaming," *IEEE Transactions on Broadcasting*, vol. 64, no. 2, pp. 474–487, Jun. 2018. [Online]. Available: https://doi.org/10.1109/TBC.2018.2822870

[8] Y. Li *et al.*, "Demystifying the qos and qoe of edge-hosted video streaming applications in the wild with sneset," *Proceedings of the ACM on Management of Data*, vol. 1, no. 4, 2023. [Online]. Available: https://doi.org/10.1145/3626723

[9] K. Nahrstedt and J. M. Smith, "The qos broker," *IEEE MultiMedia*, vol. 2, no. 1, pp. 53–67, 1995.

[10] M. Shamsabadi, J. D'Souza, and S. Auer, "Large language models for scientific information extraction: An empirical study for virology," *arXiv:2401.10040*, 2024.

[11] J. Dagdelen *et al.*, "Structured information extraction from scientific text with large language models," *Nature Communications*, vol. 15, no. 1, p. 1418, 2024. [Online]. Available: https://www.nature.com/articles/s41467-024-45563-x

[12] D. Wang *et al.*, "Docllm: A layout-aware generative language model for multimodal document understanding," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Bangkok, Thailand: Association for Computational Linguistics, Aug. 2024, pp. 8529–8548. [Online]. Available: https://aclanthology.org/2024.acl-long.463/

[13] A. Madaan *et al.*, "Self-refine: Iterative refinement with self-feedback," *arXiv:2303.17651*, 2023.

[14] J. Gu *et al.*, "A survey on llm-as-a-judge," 2024.

[15] OpenAI, "Introducing deep research," Feb. 2025, accessed: 2026-03-26. [Online]. Available: https://openai.com/index/introducing-deep-research/

[16] B. Wang *et al.*, "Mineru: An open-source solution for precise document content extraction," *arXiv:2409.18839*, 2024.

[17] OpenAI, "Introducing gpt-5.2," Dec. 2025, accessed: 2026-03-27. [Online]. Available: https://openai.com/index/introducing-gpt-5-2/

[18] Google, "Gemini 2.5 flash-lite," 2025, accessed: 2026-04-01. [Online]. Available: https://ai.google.dev/gemini-api/docs/models/gemini-2.5-flash-lite

[19] Anthropic, "Introducing claude haiku 4.5," Oct. 2025, accessed: 2026-04-01. [Online]. Available: https://www.anthropic.com/news/claude-haiku-4-5

[20] xAI, "Grok 4.20 0309 reasoning," accessed: 2026-04-01. [Online]. Available: https://docs.x.ai/developers/models/grok-4.20-0309-reasoning

[21] A. Yang *et al.*, "Qwen3 technical report," *arXiv:2505.09388*, 2025.

[22] Qwen Team, "Qwen3.5-35b-a3b," official model card. Accessed: 2026-03-27. [Online]. Available: https://huggingface.co/Qwen/Qwen3.5-35B-A3B

[23] A. Grattafiori, A. Dubey *et al.*, "The llama 3 herd of models," *arXiv:2407.21783*, 2024.

[24] Thinking Machines Lab, "Tinker," 2025, accessed: 2026-04-02. [Online]. Available: https://thinkingmachines.ai/tinker/